Vectors I

Scalar Product and Vector Product (draft sample)

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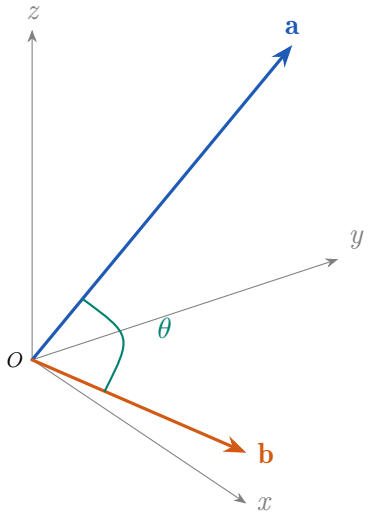
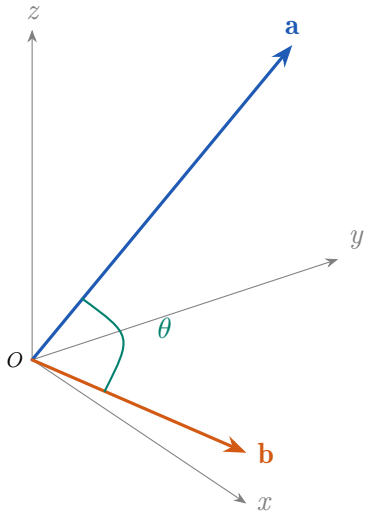
Two products, two answers



$$\mathbf{a} \cdot \mathbf{b} = \text{scalar}$$

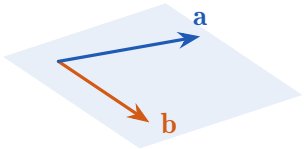
$$\mathbf{a} \times \mathbf{b} = \text{vector}$$

Scalar product

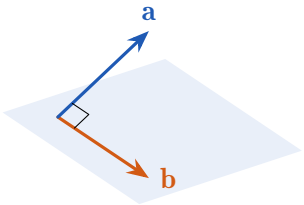


$$\begin{aligned}\mathbf{a}\cdot\mathbf{b} &= a_1b_1 + a_2b_2 + a_3b_3 \\ &= |\mathbf{a}||\mathbf{b}|\cos\theta\end{aligned}$$

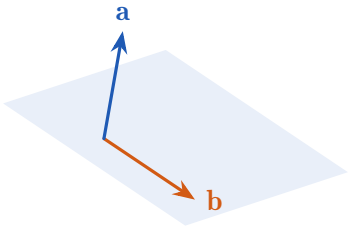
Sign of $\mathbf{a} \cdot \mathbf{b}$



> 0 acute

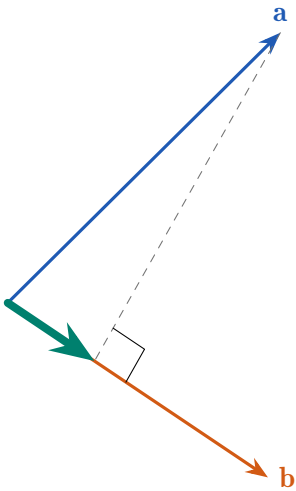


$= 0$ $\perp$



< 0 obtuse

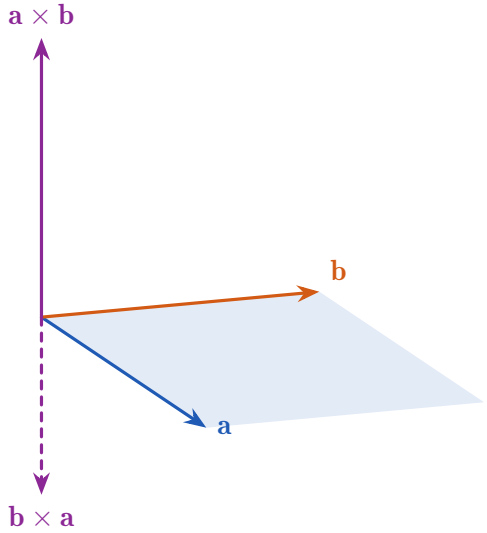
Projection of **a** onto **b**



$$\text{length} = |\mathbf{a} \cdot \hat{\mathbf{b}}|$$

$$\text{vector} = (\mathbf{a} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}}$$

Vector product: perpendicular to both

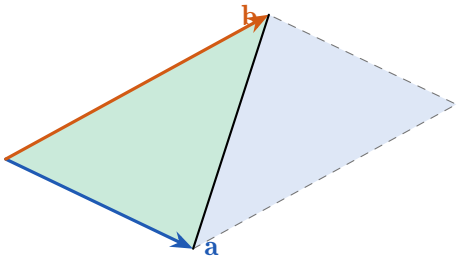


$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ -(a_1 b_3 - a_3 b_1) \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0$$

$$\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$$

Area

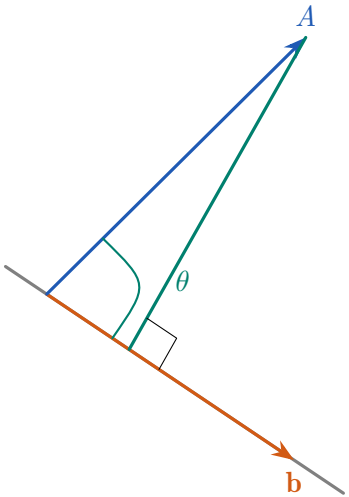


$$\text{parallelogram} = |\mathbf{a} \times \mathbf{b}|$$

$$\triangle OAB = \frac{1}{2}|\mathbf{a} \times \mathbf{b}|$$

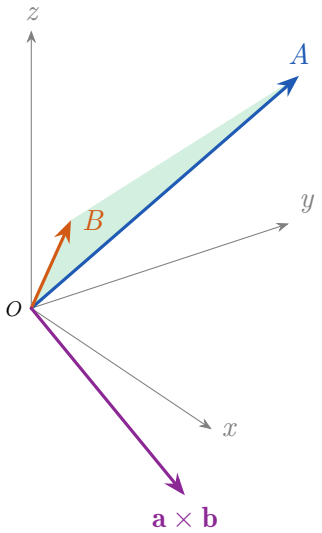
$$\triangle ABC = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$$

Perpendicular distance



$$AF = |\mathbf{a}| \sin\theta$$
$$= |\mathbf{a} \times \hat{\mathbf{b}}|$$

Example



$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 6 \\ 2 \\ -5 \end{pmatrix}$$

$$\triangle OAB = \frac{\sqrt{65}}{2}$$

Summary

	$\mathbf{a} \cdot \mathbf{b}$	$\mathbf{a} \times \mathbf{b}$
result	scalar	vector $\perp$ both
formula	$\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \cos\theta$	$\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \sin\theta$
test	$= 0 \iff \perp$	$= \mathbf{0} \iff \parallel$
use	projection, angle	area, distance